8.3 Answers: volume and angles

← Unit overview · Practice

For the parent: for the drawing questions, a result within 2° is a good result. School protractors and pencil lines are not more exact than that.

Set A: Litres and friends

1. **1 000 ml.** Milli means one thousandth.
2. **2 000 ml.** 2 × 1 000.
3. **0,5 l.** Half of 1 000 ml.
4. **1 l.** The litre cube from activity 1.
5. **1 ml.** A sugar-cube-sized space.
6. **3 l.** 1 dm^3 = 1 l.
7. **2,5 dl and 25 cl.** 1 dl = 100 ml; 1 cl = 10 ml.
8. **1 500 ml.** 1 000 + 500.
9. **1 000 dm^3 = 1 000 l.** 10 × 10 × 10 litre-cubes.
10. **750 ml = 0,75 l.** 1 cm^3 = 1 ml.
11. **10 dl.** One step on the staircase.

Set B: Volume of a box or cube

1. **1 cm^3.** 1 × 1 × 1.
2. **8 cm^3.** 2 × 2 × 2.
3. **6 cm^3.** 3 × 2 × 1.
4. **40 cm^3.** 5 × 4 × 2.
5. **1 000 cm^3 = 1 litre.** 10 × 10 × 10.
6. **6 000 cm^3 = 6 litres.** 30 × 20 × 10; 6 000 : 1 000.
7. **60 litres.** 50 × 30 × 40 = 60 000 cm^3.
8. **25 cm.** 1,5 l = 1 500 cm^3; the base is 10 × 6 = 60 cm^2; 1 500 : 60 = 25.
9. **0,6 m^3 = 600 litres.** 2 × 1,5 × 0,2 = 0,6; 1 m^3 = 1 000 l.
10. **60 cubes.** 4 × 3 = 12 in one layer, 5 layers.
11. **27 cm^3.** 3 × 3 × 3.

Set C: Kinds of angles

1. **Right angle, 90°.**
2. **Acute** (less than 90°).
3. **Obtuse** (between 90° and 180°).
4. **Straight angle.**
5. **90°.** 3 steps × 30°.
6. **180°.** 6 steps × 30°.
7. **30°.** 360 : 12.
8. **60°.** 2 × 30°.
9. **180°, 90°, 270°.** Half, quarter and three quarters of 360.
10. **45°.** Half of 90.
11. **Obtuse.**

Set D: Measure, draw and find the missing angle

1. **Drawing.** The paper corner should fit exactly.
2. **Drawing.** Should look like half a right angle.
3. **Drawing.** A third of a right angle.
4. **Drawing; obtuse.** It is a right angle plus half a right angle.

5. **50°.** 180 − 130.
6. **50°.** 180 − 90 − 40.
7. **60°.** 180 − 60 − 60. (All three are equal: an equilateral triangle, *gelijkzijdige driehoek*.)
8. **30°.** 180 − 100 − 50.
9. **120°.** 360 : 3.
10. **360°.** 4 × 90.

Puzzle corner

1. **75°.** The minute hand is on the 6 (180°). The hour hand is halfway between 3 and 4, at 3,5 × 30° = 105°. 180 − 105 = 75.
2. **1 000 cubes.** 10 × 10 × 10.
3. **8 times as much.** 2 × 2 × 2. Try it with Lego: a 1-brick cube becomes 8 bricks.